

Operational Technology Graduate Demonstration Project Vision

Project Vision Brief

Background

I recently applied for the Graduate Operational Technology Engineer position at Oribus. Rather than creating a generic software portfolio project, this project is intended to simulate the type of learning, engineering thinking and practical work a graduate might undertake while contributing to an Operational Technology (OT) consulting project.

Following research into Oribus' services, the graduate role, and the operational technology environment of electricity utilities (including Oribus' involvement in TasNetworks' Advanced Distribution Management System (ADMS) project), the aim is not to build a commercial SCADA or ADMS platform.

Instead, this project represents a **small fictional graduate work package within an electricity distribution system upgrade project**.

Project Objective

The objective is to demonstrate that I have proactively begun preparing for the type of work performed by an Operational Technology graduate consultant.

Rather than demonstrating only software development ability, the project should demonstrate:

- understanding of electricity distribution operations
- understanding of operational technology systems and their roles
- systems thinking
- structured engineering problem solving
- testing and validation methodology
- configuration and defect investigation
- technical communication and documentation
- practical software implementation

The project should answer the question:

“Would this person be able to become productive more quickly because they have already invested time understanding our industry and the way our projects operate?”

Fictional Project Scenario

A fictional electricity distribution utility is upgrading part of its operational control environment.

A simplified model of the distribution network has been created to support the upgrade.

As part of the upgrade project, a graduate engineer has been assigned responsibility for validating a small section of the upgraded operational workflow.

The work package includes:

- understanding the network model
- validating operational behaviour
- executing predefined test cases
- investigating unexpected behaviour
- identifying configuration or data issues
- documenting findings
- correcting the identified issue
- validating the corrected system

This mirrors the type of supervised engineering work a graduate consultant could realistically contribute to during an operational technology implementation project.

Demonstrator Scope

The demonstrator is **not** intended to replicate commercial operational technology platforms.

It is intended to simulate a very small engineering work package that demonstrates understanding of how multiple operational systems interact during a network event.

The demonstrator will include simplified representations of concepts such as:

- SCADA telemetry
- network topology
- feeder status
- alarms

- outage information
- operational workflows
- automation logic
- event logging
- engineering validation
- defect investigation

All network data will be fictional.

No real utility network or operational information will be recreated.

Core Engineering Scenario

The project will centre around a feeder fault occurring on a simplified electricity distribution network.

The validation process will demonstrate that:

- downstream sections become de-energised
- alarms are generated correctly
- affected customers are identified
- restoration is only proposed where electrically valid
- simplified feeder capacity constraints are respected
- uncertain or stale telemetry prevents unsafe actions
- every engineering action is traceable through system logs

One intentionally seeded configuration or data issue will cause a validation test to fail.

The project will demonstrate the engineering workflow required to:

1. detect the issue
 2. investigate the cause
 3. identify the incorrect configuration
 4. implement a correction
 5. repeat validation
 6. confirm successful behaviour
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Educational Focus

The project is equally a learning exercise and an engineering demonstration.

During development I will study and document:

- electricity fundamentals
- distribution network operation
- operational technology concepts
- SCADA
- ADMS
- OMS
- GIS
- operational workflows
- testing methodology
- engineering validation
- system integration concepts

The project documentation should clearly show how this understanding developed throughout the project.

Deliverables

The completed project will include:

- Engineering Design Brief
 - Operational Technology research
 - Requirements specification
 - Simplified network model
 - Operational workflow diagrams
 - System architecture diagrams
 - Data flow diagrams
 - Requirements traceability matrix
 - Validation test plan
 - Executed validation results
 - Defect investigation report
 - Demonstrator application
 - Technical documentation
 - Final evaluation and lessons learned
 - Short demonstration walkthrough
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Guiding Principles

This project is **not** intended to demonstrate expertise in proprietary utility software or advanced power-system engineering.

Instead, it aims to demonstrate:

- initiative
- structured learning
- engineering thinking
- systems thinking
- attention to detail
- practical implementation
- technical communication
- an understanding of how Operational Technology consulting projects are delivered.

Every feature, document and engineering decision should be traceable back to one question:

“Does this help demonstrate the kind of understanding and contribution expected from a graduate Operational Technology consultant working on a real client project?”

Project Success Criteria

The project will be considered successful if, upon reviewing it, an experienced Operational Technology consultant could reasonably conclude:

“This candidate clearly recognised the gaps in their experience, invested time understanding our domain, and approached the project like a junior engineer contributing to an operational technology upgrade rather than simply building another software application.”
